

Review

Herpes simplex virus infections

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Herpes simplex virus (HSV) is a member of the herpesviridae family. Recognised since ancient Greek times, the virus frequently infects human beings, causing a range of diseases from mild uncomplicated mucocutaneous infection to those that are life threatening. In the past 50 years, substantial advances in our knowledge of the molecular biology of HSV have led to insights into disease pathogenesis and management. This review provides a contemporary interpretation of the biological properties, function, epidemiology, and treatment of HSV diseases.

Herpes, from the ancient Greek meaning to creep or crawl, is the name of a family of viruses of which herpes simplex virus 1 and herpes simplex virus 2 (HSV-1 and HSV-2) are the most serious human pathogens. HSV-1 is normally associated with orofacial infections and encephalitis, whereas HSV-2 usually causes genital infections and can be transmitted from infected mothers to neonates. Both viruses establish latent infections in sensory neurons and, upon reactivation, cause lesions at or near point of entry into the body. This latency increases the pathogenicity of HSV and will enable these viruses to be used as therapeutic tools.

Structure

HSV-1 and HSV-2 contain a large double-stranded DNA molecule.¹ The HSV virion has four parts: an electron-dense core containing viral DNA; an icosapentahedral capsid; a tegument—an amorphous layer of proteins that surround the capsid; and an envelope. The DNA of HSV-1 and HSV-2 consists of two covalently linked components, L (long) and S (short), with unique sequences—U_L (unique long) or U_S (unique short)—flanked by large inverted repeats. The two components can invert relative to one another, creating four different types of DNA molecule that differ only in their orientation of DNA sequences.² After the structure of HSV was understood, the virus was genetically engineered and used in vaccine development and gene therapy.

HSV-1 and HSV-2 encode at least 84 different polypeptides.³ Each protein does many things, hence HSV genes can encode several hundred different functions. To initiate infection, HSV attaches to at least three different classes of cell-surface receptor and fuses its envelope with the plasma membrane. The capsid, minus its envelope, is transported to the nuclear pore, through which it releases viral DNA into the nucleus. HSV replicates by three rounds of transcription that yield: α (immediate early) proteins that mainly regulate viral replication; β (early) proteins that synthesise and package DNA; and γ (late) proteins, most of which are virion proteins.³ Of the 84 known polypeptides, at least 47 are not needed for viral replication in cultured cells. These 47 genes are not

completely dispensable. Some complement cellular genes that are not expressed in terminally differentiated cells; others alter cellular metabolism to ensure high virus yields. Mutant viruses lacking these genes cannot survive in nature.

Viral function

Several parts of the viral replication cycle and the resulting proteins are relevant to the study of human disease and antiviral therapy. For example, viral surface glycoproteins mediate attachment and penetration of HSV into cells, and provoke host immune responses. There are 11 known viral glycoproteins (B–M) and a 12th (gN) is predicted. Two of these glycoproteins (gB or gD, or both) have been used in subunit vaccines. Replication-defective vaccines are made from viruses that lack one or more essential genes.^{4,5}

At least six viral proteins ensure robust expression of viral genes and mobilise cellular proteins to synthesise viral DNA and proteins efficiently. A tegument protein, the product of the U_L41 gene, interacts with a cellular protein to degrade mRNA soon after infection. Another tegument protein, VP16, enhances expression of α genes. Infected cell proteins 0, 4, and 22 (ICP0, ICP4, and ICP22) are multifunctional viral proteins that alter the environment of the infected cell to allow efficient high-level expression of viral genes. ICP0 is expressed in both the nucleus and cytoplasm; its role in the ubiquitin-proteasomal pathway is being researched.² ICP47 is another multifunctional protein; it blocks RNA splicing early in infection, then, late in infection, shuttles viral mRNA from nucleus to cytoplasm. The regulatory proteins ICP0, ICP4, and ICP22 are phosphorylated by cellular kinases (eg, cell cyclin kinase, cdc2) and viral-protein kinases (U_S3 and U_L13). U_L13 phosphorylates many viral and cellular proteins, and is often targeted in antiviral chemotherapy.

Viral DNA synthesis requires at least seven viral proteins. Other viral proteins, mainly thymidine kinase, ribonucleotide reductase, dUTPase, and uracil DNA glucosylase, control viral nucleic acid metabolism and are current (polymerase and thymidine kinase) or potential targets of antiviral therapy. Furthermore, several HSV proteins package viral DNA into a preformed capsid; a potentially important antiviral target is the protease that cleaves scaffolding proteins after this process. Figure 1 shows maturation of the virus in a host cell.

HSV has an elaborate system for blocking host responses to infection. Cells degrade some of the newly synthesised viral proteins as well as cell proteins specifically targeted for degradation. Peptides

Lancet 2001; **357**: 1513–18

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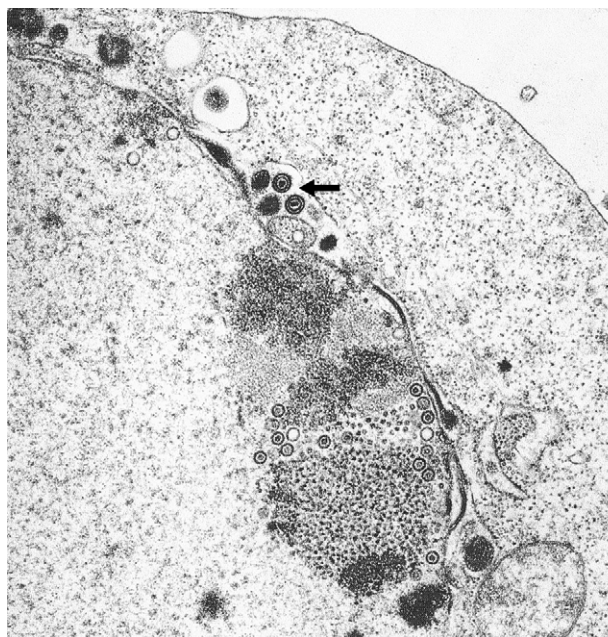


Figure 1: **Maturation of herpes simplex virus**
Black arrow=mature herpes simplex virion.

(degradation products) are transported to the endoplasmic reticulum, and those of correct size and conformation are presented by MHC-class-1 proteins. HSV blocks presentation of peptides by encoding ICP47, a small protein that binds to the transporter protein TAP1 or TAP2 and stops it from transporting peptides to the endoplasmic reticulum.³

Another host defence is programmed cell death in response to disturbance of cellular machinery by viral proteins or host immune system. HSV blocks programmed cell death whether caused by its own or by host proteins. At least three proteins, U_s3 , gJ, and gD, are thought to block programmed cell death caused by specific cell injury. Finally, HSV, like most viruses, makes complementary RNA that can anneal to and activate host protein kinase R. This enzyme can phosphorylate the translation initiation factor eIF-2 α , which in turn blocks all protein synthesis. HSV has a counter defence; HSV protein $\gamma_134.5$ binds protein phosphatase 1 and directs it to dephosphorylate eIF-2 α with great efficiency.³ Also, cells cannot respond to infection because HSV destroys cellular mRNA, inhibits DNA transcription, and blocks splicing of mRNA.

Biological properties

The biological properties of HSV that control the course of human-HSV infection are: neuroinvasiveness—the ability to invade the brain; neurotoxicity—the ability to multiply and destroy the brain; and latency—the ability to remain in a non-replicating form in neurons of dorsal root ganglia and the autonomic nervous system. Although neuroinvasiveness and neurotoxicity seem to be distinct characteristics, deletion of almost any HSV gene affects one or both properties but not the ability to establish latency. For example, mutations in genes encoding glycoproteins can alter neuroinvasiveness. Furthermore, non-dividing cells, such as neurons, do not make enzymes for deoxynucleotide synthesis and, hence, do not support replication of viral mutants lacking these genes. Lack of neurotoxicity has been mapped to several domains of the HSV genome but, in particular, to the inverted repeats flanking the L and S components of viral DNA. $\gamma_134.5$

deletion mutants have low neurotoxicity even though they multiply well in cell culture.⁶ The mechanism of how $\gamma_134.5$ affects viral growth in the brain is not known but is probably different from how it directs dephosphorylation of eIF-2 α .

HSV-1 and HSV-2 are transported by retrograde flow along axons that connect the point of entry into the body to nuclei of sensory neurons.⁷ Viral multiplication occurs in a small number of sensory neurons; the viral genome then remains in a latent state for the life of the host. Many events cause viral reactivation, such as physical or emotional stress, fever, ultraviolet light, and tissue damage. Reactivation can occur even in hosts with cell-mediated and humoral HSV immunity. Recurrent herpes labialis is three times more common in febrile than non-febrile patients. Latent virus can be reactivated from the trigeminal, sacral, and vagal ganglia of human beings either unilaterally or bilaterally in cell culture. Viral-gene expression is thought to be required for reactivation but not for establishment of latency. How latency is established is not known. The leading model is that a cellular protein, C1, is needed for viral-gene expression—hence absence of this protein causes latency—and that unidentified cellular repressors block basal-level gene expression.

Epidemiology

HSV must contact mucosal surfaces or abraded skin to initiate infection. The type of HSV infection that results is a function of the host's immune status. Susceptible individuals (ie, those who are HSV seronegative) develop "primary" infection after their first exposure to HSV-1 or HSV-2. "Initial" infection occurs when an individual who has antibodies to either HSV-1 or HSV-2 is infected with the other virus type for the first time. HSV-1 and HSV-2 are usually transmitted by different routes and affect different areas of the body, but the signs and symptoms that they cause overlap.⁸ These viruses occur worldwide, have no seasonal variation, and naturally only infect human beings. HSV infection is rarely fatal. Most human beings have been infected and harbour latent virus that can reactivate; hence there is a vast HSV reservoir for transmission to susceptible individuals.

Many demographic factors affect acquisition of HSV-1 infection. In less developed countries seroconversion happens early in life—at 5 years in around a third of children, and in around 70–80% by adolescence. In comparison, middle-class and upper-class individuals in more developed countries become infected later on—seroconversion occurs in about 20% of children younger than 5 years; then no substantial rise in incidence happens until an increase to 40–60% at age 20–40 years. In the USA, race also affects acquisition of HSV-1. By age 5 years, more than 35% of African-American versus 18% of white children are infected with HSV-1. Incidence of infection among university students is around 5–10% annually.⁹

HSV-2 infections are usually sexually transmitted. Most genital HSV infections are caused by HSV-2; however, an increasing proportion is attributable to HSV-1. Genital HSV-1 infections are usually less severe and less prone to recur than those caused by HSV-2. HSV-2 seroprevalence rises from about 20–30% at age 15–29 years to 35–60% by age 60 years. This prevalence increased dramatically from the late 1970s to late 1980s.^{9,10} Factors that affect acquisition of HSV-2 infection include sex (infection is more frequent in women), race (infection is more frequent in African Americans than whites), marital status, number of sexual partners, and place of residence (prevalence is



Figure 2: **Recurrent herpes simplex labialis**

higher in cities than suburbs). Prevalence of antibodies to HSV-2 is highest among female prostitutes (75%) and male homosexuals (83%).⁹

As with HSV-1 infection of the mouth, HSV-2 primary, initial, or recurrent infection can be symptomless. Recurrence varies between men and women, occurring 2·7 and 1·9 times per 100 days, respectively. Women with initial genital herpes can shed the infection without symptoms; this occurs in 12%, 18%, and 23% of primary HSV-1, primary HSV-2, and non-primary-HSV-2 infections, respectively. In cells cultured from women with established genital-HSV-2 infection, symptomless shedding occurs on 3–5% of all days cultured. Analysis of swabs by PCR increases this estimation of rate to 28%, which suggests that infection is chronic rather than intermittent.¹¹

Cell characteristics of primary or recurrent HSV infection are those of viral-mediated cell death and associated inflammatory response. Viral infection causes cells (usually parabasal and intermediate cells of the epithelium) to swell with condensed chromatin in the nucleus, which is followed by nuclear degeneration. Plasma membranes break and infected cells form multinucleated giant cells.

Diagnosis

Diagnosis can be confirmed by virus isolation in cell culture or PCR detection of HSV DNA. Cell culture is less expensive than PCR¹² and provides a virus isolate that can be typed. A scraping of skin vesicles should be taken from any skin lesions and transferred in appropriate virus transport media, preferably on ice, to a diagnostic virology laboratory for inoculation onto a susceptible cell culture. Cytopathic effects usually develop 24–48 h after inoculation. Serological diagnosis of HSV infection is of value only to determine past exposure. Therapeutic decisions cannot await findings of serological studies. Serological assays that distinguish between antibodies to HSV-1 and HSV-2 are only now being licensed by worldwide regulatory authorities. PCR analysis of cerebrospinal fluid for HSV DNA is the preferred diagnostic method for HSV brain infections. Cerebrospinal fluid will stay PCR positive for up to 1 week in most patients with such infections.

Signs and symptoms

The most common sites of HSV infection are skin and mucosal membranes, irrespective of virus type. The

incubation period for HSV-1 or HSV-2 is about 4 days, and ranges from 2 to 12 days. Most people do not notice an HSV-1 or HSV-2 infection.⁸ Primary HSV-1 infection of the oropharynx and HSV-2 infection of the genital tract result in virus excretion for up to 23 days (average of around 7–10 days). Symptomatic oropharyngeal disease is characterised by lesions of the buccal and gingival mucosa (lasting 2–3 weeks), and by fever between 38·3 and 40° C. Intraoral ulceration lesions indicate primary infection, whereas lip lesions suggest recurrent infection.

Recurrent orolabial lesions are preceded by pain, burning, tingling, or itching, which usually lasts for 6 h; vesicles then appear on the vermillion border of the lip (figure 2).¹³ Three to five vesicles normally occur, which persist for about 48 h, then become pustules or ulcers and then form crusts within 72–96 h; lesions are completely healed after 8–10 days. Pain is most severe at the start of infection, then resolves over 96–120 h. The frequency of recurrence varies among individuals. Other skin HSV-1 infections can occur, such as: eczema herpeticum in patients with underlying atopic dermatitis; herpes gladiatorum in wrestlers; and extensive lesions associated with Darier's and Sézary's syndromes. HSV infections of any type can trigger erythema multiforme.

Primary genital herpes appears as macules and papules, followed by vesicles, pustules, and ulcers.^{14,15} Complications in men are rare; aseptic meningitis and urinary retention are more common in women. Other complications in men and women include paraesthesias and dysaesthesias of the legs and perineum, dysuria, localised inguinal adenopathy, and malaise. Primary perianal HSV-2 infections and proctitis are most common in male homosexuals. Non-primary initial genital infection causes milder symptoms than primary infection (fewer lesions, less pain, and less likelihood of complications), and heals in around 2 weeks. Antibodies to HSV-1 reduce disease severity of HSV-2.

Recurrent male genital infection appears as three to five vesicles on the shaft of the penis; female infection causes ulcerating-vesicle genital lesions or merely vulvar irritation. The disease lasts around 8–10 days. A third of patients are estimated to have more than six recurrences per year, a third will have two per year, and the remaining third will have rare recurrences. HSV can be transmitted to sexual partners in the presence or absence of symptoms.¹⁶ Indeed, transmission usually occurs from symptomless virus shedding. Wald and colleagues¹⁷ showed a three-fold higher frequency of HSV DNA shedding (detected by PCR) in genital secretions between, rather than during, clinical recurrences.

Genital-HSV infection is rarely transferred from mother to fetus during pregnancy; when it is transferred, it involves multiple visceral sites, and causes necrotising hepatitis with or without thrombocytopenia, disseminated intravascular coagulopathy, and encephalitis. Fetal infection occurs in about one in 300 000 live neonates in North America. It is associated with microcephaly, skin scarring, and retinitis.

Neonatal HSV has an incidence of around one in 2000 to one in 5000 births per year in the USA, but the incidence is substantially lower in the UK. In any country, this incidence seems to be directly related to seroprevalence of HSV-2. At least three factors affect transmission of infection from mother to fetus. Maternal primary or initial infection is transmitted to 30–50% of vaginally delivered babies—higher than the recurrent infection transmission frequency of 3% or less.¹⁸ Transplacental maternal antibodies possibly reduce severity of infection if not actually prevent transmission,

Indications for aciclovir therapy among patients with herpes simplex virus infection

Type of infection	Route and dose*	Comments
Genital HSV		
Initial episode		
Aciclovir	200 mg by mouth 5 times/day for 7–10 days 5 mg/kg intravenous every 8 h for 5–7 days 400 mg by mouth 3 times/day	Preferred route in normal host Severe cases only
Valaciclovir	1 g by mouth twice daily for 7–10 days	
Famciclovir	250 mg by mouth 3 times, ie 3 times/day for 5–10 days	
Recurrent episode		
Aciclovir	400 mg by mouth twice daily for 5 days	Limited clinical benefit
Valaciclovir	500 mg by mouth twice daily for 5 days	
Famciclovir	125–250 mg by mouth twice daily for 5 days	
Suppression		
Aciclovir	400 mg by mouth twice daily	Titrate dose as required
Valaciclovir	500 or 1000 mg by mouth once daily	
Famciclovir	250 mg by mouth twice daily	
Mucocutaneous HSV in immunocompromised patients		
Aciclovir	200–400 mg by mouth 5 times/day for 10 days 5 mg/kg intravenously every 8 h for 7–14 days 400 mg by mouth 5 times/day for 7–14 days	For minor lesions only
Valaciclovir	500 mg twice daily by mouth	
Famciclovir	250 mg 3 times/day by mouth	
HSV encephalitis		
Aciclovir	10–15 mg/kg intravenously every 8 h for 14–21 days	
Neonatal HSV†		
Aciclovir	20 mg/kg intravenously every 8 h for 14–21 days	

*Doses are for adults with normal renal function unless otherwise noted. †Not currently approved by the US Food and Drug Administration. HSV=herpes simplex virus.

and membrane rupture that lasts for more than 6 h increases the risk of infection. Fetal scalp electrodes can also be a site for HSV transmission. Neonate infection can occur in utero (about 5% of infections), intrapartum (around 80%), or postnatally. Infection is nearly always symptomatic and frequently lethal. Disease can occur locally in: skin, eye, or mouth (40% of babies); cause encephalitis with or without skin infection (35%); or result in disseminated infection (25%).⁸

Around 300 000 cases of HSV eye infections are diagnosed yearly in the USA. Herpetic keratoconjunctivitis is associated with unilateral or bilateral conjunctivitis, which can be follicular, followed soon by preauricular adenopathy. Geographic ulcers of the cornea develop with advanced disease. Recurrences are common. Immunocompromised patients, especially organ transplant recipients, are at increased risk of severe HSV infection. These patients can develop progressive disease involving respiratory tract, oesophagus, or gastrointestinal tract.

HSV encephalitis is a devastating disease and is thought to be the most common cause of sporadic fatal encephalitis, which is usually focal. More than 70% of untreated patients die and only 2.5% of surviving patients return to normal neurological function.⁸ HSV can also affect almost all areas of the nervous system, causing meningitis, myelitis, and radiculitis, among other diseases. HSV has been isolated from the respiratory tract of adults with adult respiratory distress syndrome and acute onset bronchospasm; it increases the rates of mortality and morbidity of these problems.

Treatment

Drugs

Aciclovir, a synthetic acyclic purine-nucleoside analogue, is the standard therapy for HSV infections¹⁹ and has greatly helped control symptoms. Precursor drugs,

valaciclovir (converted to aciclovir) and famciclovir (converted to penciclovir), have been licensed and have better oral bioavailability than aciclovir and penciclovir, respectively.²⁰

Mucous membrane HSV infections

Initial genital or oral HSV infection can be treated with topical, oral, or intravenous aciclovir. Topical therapy is much less effective than oral or intravenous treatment. Although intravenous aciclovir is the most effective treatment for first-episode genital herpes, oral aciclovir therapy of 200 mg, five times daily, is preferred (panel). Valaciclovir and famciclovir have similar benefits to aciclovir. Neither intravenous nor oral aciclovir treatment of acute HSV infection reduces frequency of recurrences. Gingivostomatitis in children can be treated with the oral suspension of aciclovir.

In recurrent genital herpes, oral aciclovir shortens duration of viral shedding and healing time (6 days treated *vs* 7 untreated) when started within 24 h of onset; time until recurrence is not affected.²¹ Valaciclovir can be taken once daily whereas famciclovir must be given twice daily.^{22,23} Only severe recurrences of labial herpes should be treated. Topical therapy with penciclovir will accelerate healing by about 1 day. Oral aciclovir reduces time to loss of crusts from around 7 to 8 days, but does not alter duration of pain or time of complete healing.

Oral aciclovir, valaciclovir, or famciclovir suppress genital herpes in patients who have frequent recurrences. Daily aciclovir reduces frequency of recurrences by up to 80%, and prevents recurrence in 25–30% of patients. Titration of the dose of aciclovir establishes the minimum dose that is effective and economical.^{24,25} Symptomless virus shedding can occur despite clinically effective suppression with aciclovir; hence person-to-person transmission is still possible.

Intravenous or oral aciclovir therapy benefits

immunocompromised hosts. Aciclovir prophylaxis in immunocompromised patients, especially those having induction chemotherapy or transplantation, reduces the rate of symptomatic HSV infection from about 70% to 5–20%;²⁶ famciclovir and valaciclovir have similar effects. Various oral dose regimens—ranging from 200 mg three times daily, to 800 mg twice daily—have been used successfully. Among bone-marrow-transplant recipients and patients with AIDS, aciclovir-resistant HSV isolates occur more frequently after therapeutic aciclovir than during prophylaxis. Aciclovir-resistant HSV isolates are usually also resistant to famciclovir or penciclovir, or both. Aciclovir is the main treatment for HSV infections in immunocompromised patients.

Other HSV infections

Intravenous aciclovir therapy reduces deaths from HSV encephalitis at 3 months from 70% to 19%; 38% of treated patients regain normal neurological function.²⁷ This therapy is also used to treat neonatal HSV infection. In localised neonatal skin disease, mortality is 0%; in encephalitis, 5%; and for disseminated disease, 25%. Morbidity varies with type of HSV-induced disease. 95% of babies with skin disease develop normally 2 years after infection, compared with 40% and 60% for babies with encephalitis and disseminated disease, respectively.²⁸ However, babies with disease apparently localised to the skin have tested PCR positive for HSV DNA in their cerebrospinal fluid—despite negative results of biochemical HSV tests. Positive HSV PCR test results after completion of therapy are correlated with neurological impairment.²⁹

Aciclovir has been successfully used to treat other HSV infections such as hepatitis, pulmonary infection, herpetic oesophagitis, proctitis, eczema herpeticum, erythema multiforme, and herpetic whitlow. HSV eye infections can be effectively treated with aciclovir or trifluridine.

Viral resistance

HSV can develop resistance to aciclovir through mutations in the viral gene that encodes thymidine kinase—by generation of thymidine-kinase-deficient mutants or by selection of mutants with a thymidine kinase unable to phosphorylate aciclovir. Most clinical HSV isolates resistant to aciclovir are deficient in thymidine kinase, although altered DNA polymerase has been detected in some. Drug resistance was thought rare, and resistant isolates were thought less pathogenic, until aciclovir-resistant HSV isolates were characterised from patients with AIDS.³⁰ These resistant mutants were thymidine-kinase deficient. Aciclovir-resistant HSV isolates have been identified as the cause of pneumonia, encephalitis, oesophagitis, and mucocutaneous infections in immunocompromised patients.

Toxicity

Aciclovir, valaciclovir, and famciclovir therapies have few adverse effects. Renal dysfunction can occur, especially in patients given large doses of aciclovir by rapid intravenous infusion, but is rare and usually reversible. Risk of nephrotoxicity can be reduced by slow infusion of aciclovir and by ensuring adequate hydration. Oral aciclovir therapy, even at doses of 800 mg five times daily, does not cause renal dysfunction. Intravenous aciclovir has been linked with disturbances of the central nervous system, which include agitation, hallucinations, disorientation, tremors, and myoclonus. Continuous aciclovir therapy has not caused long-term toxicity in the past decade.

HSV vaccines

Because HSV recurs in individuals with humoral and cell-mediated immunity, prophylactic vaccination might be difficult to develop.³¹ Two subunit vaccines have been studied for prevention and therapy. Results from a trial in which gB and gD were the vaccine subunits did not show any prophylactic or therapeutic effect.³² In another gD vaccine study, vaccination seemed to protect women but not men.³³

Alternative vaccine possibilities include: an attenuated HSV construct (γ 34.5 deletion); a disabled infectious single-cycle vaccine; or naked DNA. Those last two potential vaccines are in phase I studies. Use of such constructs requires the ability to tame a virus capable of inducing severe morbidity and mortality if it infects the brain.

Gene therapy

Genetically-engineered HSVs have undergone extensive preclinical studies and are now in clinical tests to assess their oncolytic effects in gene therapy. Viruses lacking the γ 34.5 gene cannot replicate in post-mitotic brain cells. Such viruses can be inoculated into tumours in patients with glioblastoma multiforme without inducing encephalitis.^{34,35} Viruses lacking γ 34.5 and other genes are being tested in preclinical studies as vectors for foreign-gene expression. Other useful mutants for cancer therapy lack the internal inverted repeats flanking U_L and U_S. Mutants with more extensive deletions of essential genes are being developed to serve as vectors to deliver missing or damaged cellular genes to the central nervous system. Such mutants could establish latency and express the cellular gene for a long time. HSV-gene therapy heralds a new era of HSV biology and the conversion of a hazardous foe into a user-friendly surgical tool.

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